

- "Quadrature" : approximating integrals numerically

$$I(f) = \int_a^b f(x) dx \simeq I^H(f)$$

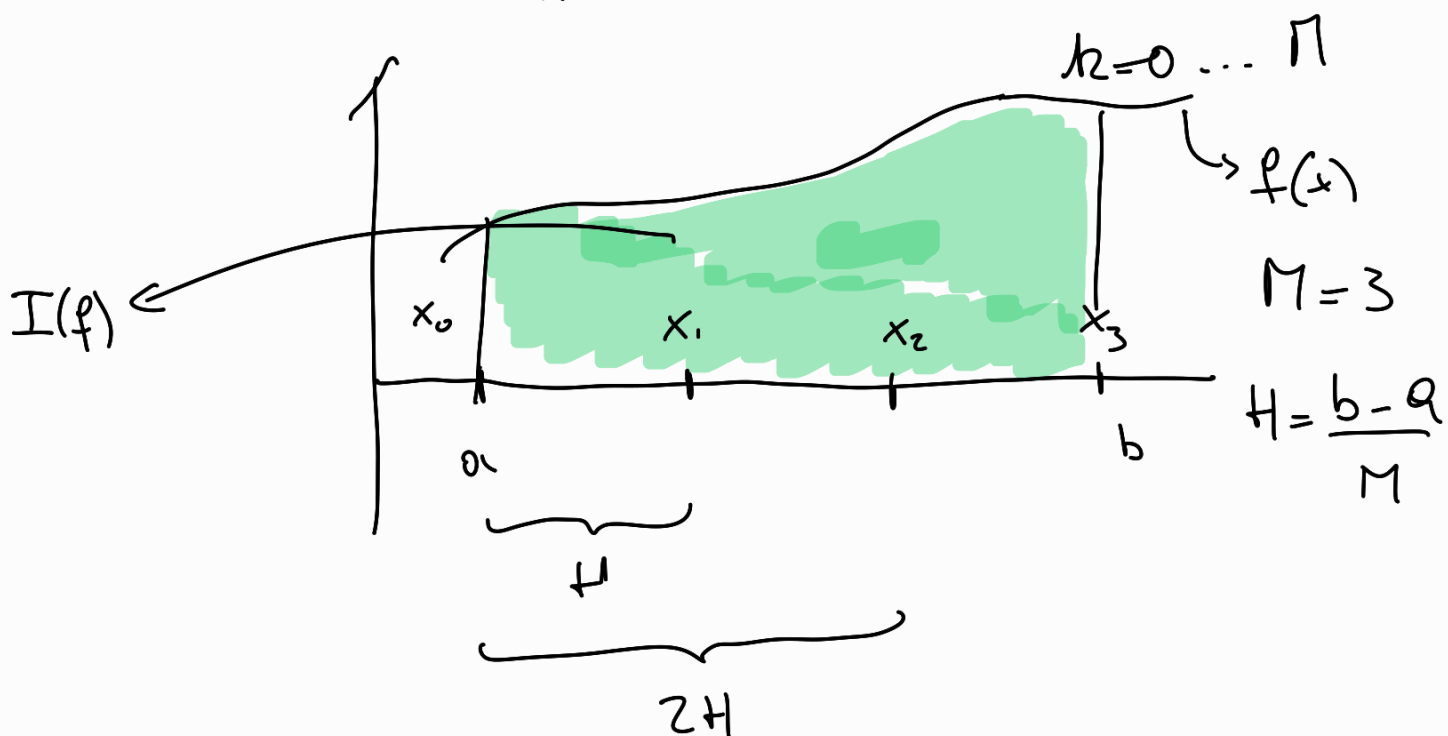
$$E_H = I(f) - I^H(f) \quad \text{Quadrature error}$$

if $E_H = 0 \quad \forall f \in \mathbb{P}_r(a, b) \rightarrow$ Degree of exactness r

$$E_H = |I(f) - I^H(f)| \leq C H^p \rightarrow \text{order } p$$

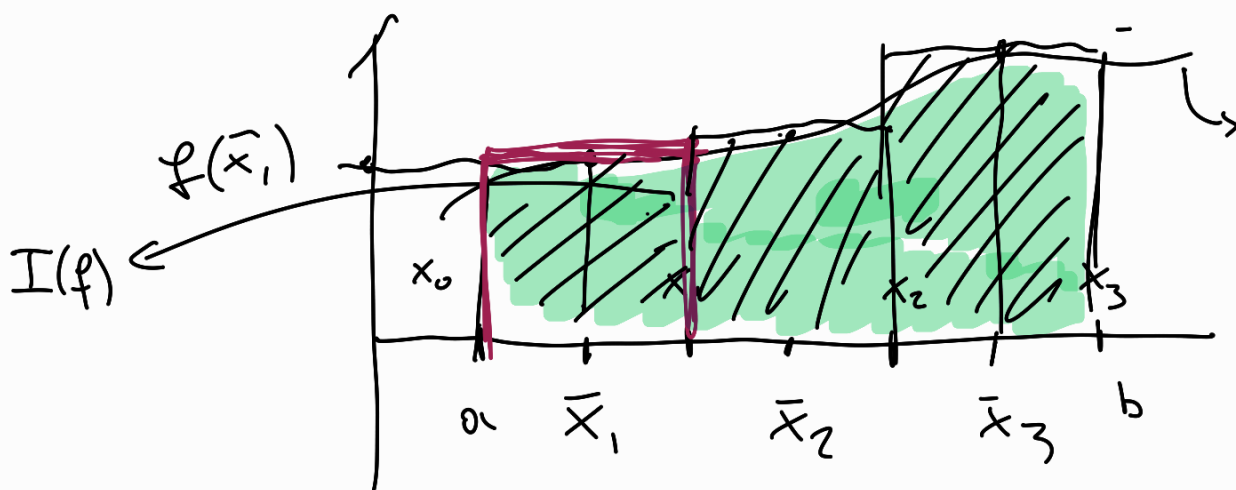
Additive property of integrals

$$[a, b] \equiv \bigcup_{k=1}^M [x_{k-1}, x_k] \quad x_k = a + kH$$



$$I(f) = \int_a^b f(x) dx = \sum_{k=1}^M \int_{x_{k-1}}^{x_k} f(x) dx$$

N.B. No approximation until now!



Midpoint rule

$$\bar{x}_k = \frac{x_{k-1} + x_k}{2}$$

$$I_{mp}^H(f) = \sum_{k=1}^M H \cdot f(\bar{x}_k) = H \sum_{k=1}^M f(\bar{x}_k)$$

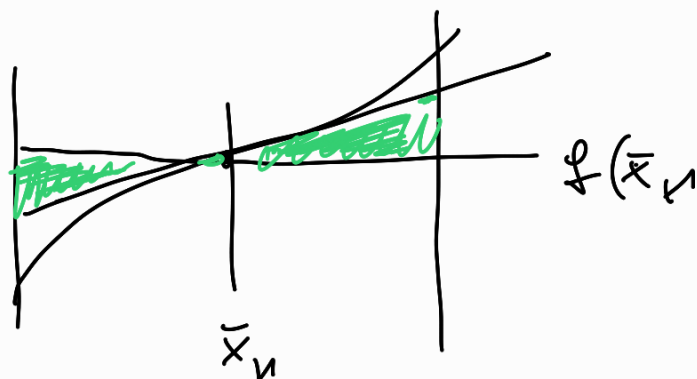
$$E_H = \int_a^b f(x) dx - I_{mp}^H(f) =$$

$$\sum_{k=1}^M \int_{x_{k-1}}^{x_k} f(x) dx - H f(\bar{x}_k)$$

$$\begin{aligned}
 & \int_{x_{k-1}}^{x_k} f(x) dx - H f(\bar{x}_k) \\
 &= \int_{x_{k-1}}^{x_k} f(x) dx - \int_{x_{k-1}}^{x_k} f(\bar{x}_k) dx \\
 &= \int_{x_{k-1}}^{x_k} (f(x) - f(\bar{x}_k)) dx.
 \end{aligned}$$

$$f(x) = f(\bar{x}_k) + f'(\bar{x}_k)(x - \bar{x}_k) + \frac{f''(\eta_k)}{2} (x - \bar{x}_k)^2$$

$$= \int_{x_{k-1}}^{x_k} f'(\bar{x}_k)(x - \bar{x}_k) dx + \int_{x_{k-1}}^{x_k} \frac{f''(\eta_k)}{2} (x - \bar{x}_k)^2 dx$$



$$\exists \xi_k \text{ in } (x_{k-1}, x_{k+1}) \text{ s.t.}$$

$$\int_{x_{k-1}}^{x_k} \frac{f''(\eta_k)}{2} (x - \bar{x}_k)^2 dx =$$

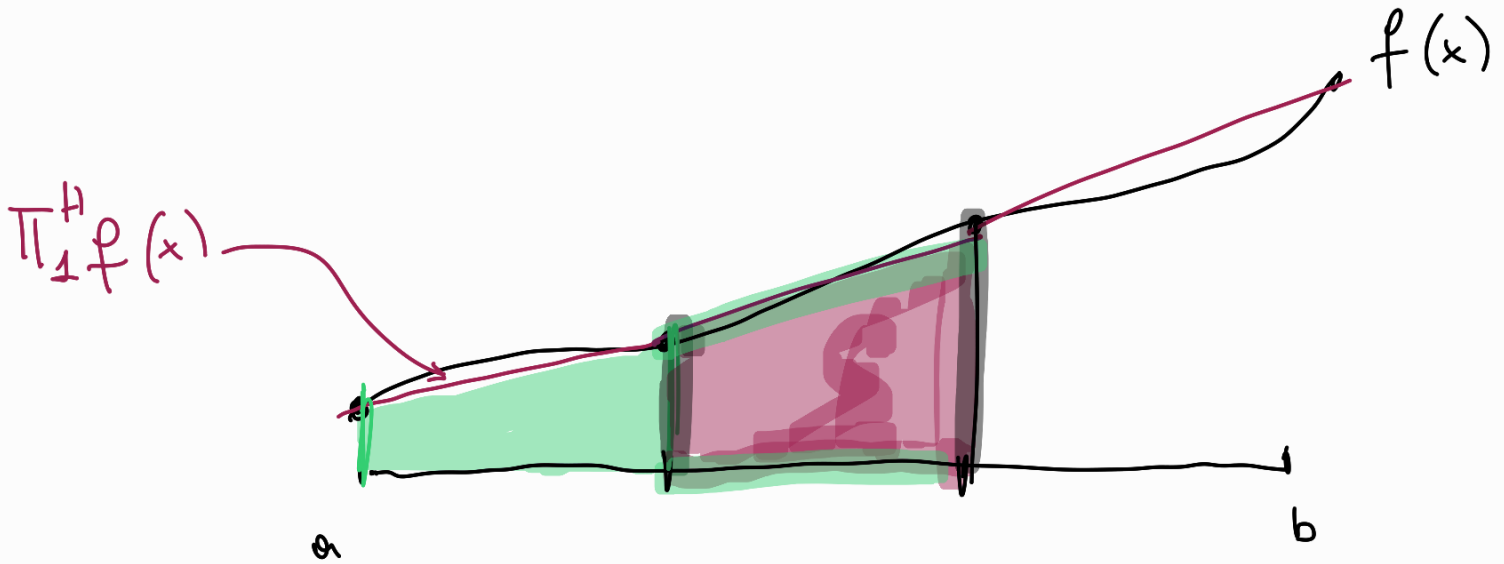
$$\frac{f''(\xi_k)}{2} \int_{x_{k-1}}^{x_k} (x - \bar{x}_k)^2 dx = \frac{f''(\xi_k)}{2} \frac{H^3}{24}$$

$$E_{pm}^4 \leq \sum_{k=1}^M \left| \frac{f''(\xi_k)}{2} \right| \frac{H^3}{24} \leq \max_{x \in [a, b]} |f''(x)| M \frac{H^3}{24}$$

$$\approx \max_{x \in [a, b]} |f''(x)| \frac{b-a}{H} \frac{H^3}{24} = \max_{x \in [a, b]} |f''(x)| (b-a) \frac{H^2}{24}$$

order 2 . exactness 1

Trapezoidal rule



$$I_{\epsilon_2}^H f = \int_a^b \Pi_1^H f(x) dx = \sum_{k=1}^M H \frac{f(x_{k-1}) + f(x_k)}{2}$$

$$H \sum_{k=1}^M \frac{f(x_{k-1}) + f(x_k)}{2}$$

$$E_{\epsilon_2}^H = \int_a^b (f(x) - \Pi_1^H f(x)) dx$$

$$= \sum_{k=1}^M \int_{x_{k-1}}^{x_k} (f(x) - \Pi_1^H f(x)) dx$$

$$= \sum_{k=1}^M \int_{x_{k-1}}^{x_k} -\frac{f''(\xi_k)}{2} (x-x_{k-1})(x-x_k) dx$$

$$= \sum_{k=1}^M -\frac{f''(\eta_k)}{12} H^3$$

$$|E_{t_2}^H| \leq \frac{b-a}{H} \max_x \frac{|f''(x)|}{12} H^3$$

$$\leq \max_x |f''(x)| \frac{H^2}{12} (b-a)$$

order 2

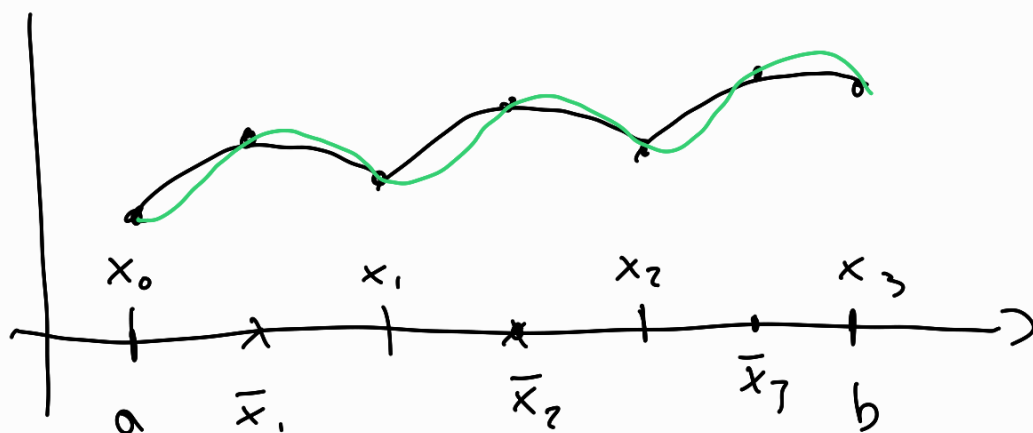
exactness 1

$$I_{t_2}^H(f) = \sum_{k=1}^M \frac{H}{2} (f(x_{k-1}) + f(x_k))$$

$$= \frac{H}{2} (f(a) + f(b)) + \sum_{k=1}^{M-1} f(x_k) H$$

Simpson's rule

$$I_s^H(f) = \int_a^b \pi_2^H(f) dx =$$



$$= \frac{H}{6} \sum_{k=1}^H \left(f(x_{k-1}) + 4f(\bar{x}_k) + f(x_k) \right)$$

N.B.

$$I_H(f) = \sum_{m=1}^n f(x_m) \omega_m H \quad \text{generic rule}$$

consistency requires that $\sum_{m=1}^n \omega_m H = (b-a)$

$$|E_s^H(f)| \leq \frac{b-a}{7880} \|f^{(4)}(x)\|_{\infty} H^4$$

order 4

exactness 3

Numerical differentiation

$$f(x) \in C^1(a, b) \quad \bar{x} \in (a, b)$$

$$f'(\bar{x}) = \lim_{h \rightarrow 0} \frac{f(\bar{x}+h) - f(\bar{x})}{h}$$

$$\approx \frac{f(\bar{x}+h) - f(\bar{x})}{h} =: d_h^+ f(\bar{x})$$

forward finite difference formula

$$d_h^- f(\bar{x}) := \frac{f(\bar{x}) - f(\bar{x}-h)}{h}$$

$$d_a^c f(\bar{x}) := \frac{f(\bar{x}+h) - f(\bar{x}-h)}{2h}$$

Convergence

$$f'(\bar{x}) - d_h^+ f(\bar{x})$$

$$f(\bar{x}+h) = f(\bar{x}) + h f'(\bar{x}) + \frac{h^2}{2} f''(\xi)$$

$$d_h^+ f(\bar{x}) = f'(\bar{x}) + \frac{h}{2} f''(\xi)$$

$$|d_h^+ f(\bar{x}) - f'(\bar{x})| \leq \frac{h}{2} \|f''\|_\infty \xrightarrow{h \rightarrow 0} 0$$

converges with order 1

$$f(\bar{x} - h) = f(\bar{x}) - h f'(\bar{x}) + \frac{h^2}{2} f''(\eta)$$

also the backward formula has order 1

$$f(\bar{x} + h) = \cancel{f(\bar{x})} + h f'(\bar{x}) + \cancel{\frac{h^2}{2} f''(\bar{x})} + \frac{h^3}{6} f'''(\xi)$$

$$f(\bar{x} - h) = \cancel{f(\bar{x})} - h f'(\bar{x}) + \cancel{\frac{h^2}{2} f''(\bar{x})} - \frac{h^3}{6} f'''(\eta)$$

$$\frac{f(\bar{x} + h) - f(\bar{x} - h)}{2h} = \cancel{2h f'(\bar{x})} + \frac{h^3}{62} (f'''(\xi) + f'''(\eta))$$

order 2